

Adaptive Hedge

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Machine learning and statistical learning — Theory Project

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A.Y. 2025/26

Paper: T. van Erven, P. Grünwald, W. M. Koolen, S. de Rooij, *Adaptive Hedge*, NeurIPS 2011
([arXiv:1110.6416](https://arxiv.org/abs/1110.6416))

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What this report proves. The paper [1] has two halves. The first half shows the worst-case scenario, where AdaHedge keeps the optimal $O(\sqrt{L_T^* \ln K})$ regret. This part rewrites standard techniques carefully, and the paper proves it in the main text. The second half shows that on easy data, when there is a clear best action, the regret is instead bounded by a constant that does not depend on the time horizon. This second half is the new contribution, and the paper moves all of its proofs to the supplementary material.

This report rebuilds that second half completely. It uses one notation throughout and fills in the missing steps. Theorem 19 below is proved step by step using a chain of five smaller results. Then the chain is tested with experiments. The worst-case half is included as well, because the easy-data argument reuses two of its lemmas. The main argument of the report is this chain:

$$w_t^* \rightarrow 1 \text{ fast} \implies \Delta_T \text{ bounded} \implies \text{the budget is never depleted} \implies m \text{ small} \implies R(T) \text{ bounded},$$

where each arrow represents a separate lemma. Section 4 measures all four arrows independently, rather than just the final result, and then discusses what the theorem does not explain.

1 Introduction and description of the problem

Decision-theoretic online learning (DTOL) [5] is the following repeated game. A set of K actions is fixed in advance. At each round $t = 1, \dots, T$ the learner publishes a probability vector $\mathbf{w}_t$ over the actions. The environment, which might be adversarial and can see $\mathbf{w}_t$, then reveals a loss vector $\boldsymbol{\ell}_t \in [0, 1]^K$. The learner pays $\mathbf{w}_t \cdot \boldsymbol{\ell}_t$. Performance is measured by the regret against the best single action in hindsight:

$$R(T) = \sum_{t=1}^T \mathbf{w}_t \cdot \boldsymbol{\ell}_t - L_T^*, \quad L_T^* = \min_{1 \leq k \leq K} L_T^k, \quad L_T^k = \sum_{t=1}^T \ell_t^k.$$

We make no probability assumptions about $\boldsymbol{\ell}_1, \dots, \boldsymbol{\ell}_T$. The guarantee must hold for every possible sequence. This is what makes the setting different from both the Bayesian and the frequentist approach.

The standard algorithm is Hedge, which plays $w_{t+1}^k \propto e^{-\eta L_t^k}$ for a parameter $\eta > 0$ called the learning rate. Everything about its performance depends on η . The problem is that the best value of η depends on the data, and we cannot know this before the data arrives.

The trade-off between the two terms. Section 2 derives the exact equation:

$$R(T) \leq \frac{\ln K}{\eta} + \Delta_T(\eta),$$

in which the two terms move in opposite directions. The first term is the cost of not knowing which action is best, and it gets smaller as η grows. The second term is the cumulative *mixability gap*. It is the cost of trusting a guess too fast, and it gets bigger as η grows.

The standard choices are $\eta = \sqrt{2 \ln K / T}$ and $\eta^* = \sqrt{2 \ln K / L_T^*}$, the second from equation (9) of the paper. We write η^* for it below. Both balance the two terms, but they do so under one assumption: that $\Delta_T(\eta)$ is as large as it can be, namely $\eta T / 8$.

That assumption gives the optimal $\Theta(\sqrt{T \ln K})$ worst-case rate. But it also makes these choices very wrong when the data is not fully adversarial. If one action is clearly the best, $\Delta_T(\eta)$ hardly grows at all.

The balance is then set at the wrong place, and the learner pays a large $\ln K/\eta$ cost for being careful when it did not need to be.

The design of AdaHedge. AdaHedge does not limit $\Delta_T(\eta)$ before seeing the data. We can observe this quantity during the run: after round t the learner knows $\delta_1, \dots, \delta_t$ exactly, because they are calculated from the weights it played and the losses it saw. So AdaHedge sets a budget $b(\eta)$. It runs at rate η until the observed $\Delta_t(\eta)$ uses up that budget, and only then lowers η .

On difficult data the budget is used up quickly, η falls, and the algorithm acts like the standard worst-case method. On easy data the budget is never used up, η never falls, and the learner keeps the aggressive rate that works well for easy data. No parameter is tuned manually, and we do not need to know T or L_T^* .

Scope of the result proved here. Proving that the budget rule is safe is the easier part: it is Theorem 13 below, and the argument uses the doubling trick together with a bound on Δ_T . The main claim is the other one: on easy data the budget is never used up at all.

For this we need a bound on δ_t that reacts to how easy the data really is, not to how bad it could be. The paper finds one. It shows that δ_t is small whenever the probability weight on the leading action is close to one (Lemma 15).

To turn that into a statement about the whole run, we have to control how fast the weight concentrates (Lemma 16). And for that we need an assumption about how fast the cumulative losses separate. Section 3 does all of this in full.

2 Notation and relevant definitions

Actions are indexed by $k \in \{1, \dots, K\}$. Cumulative losses are $L_t^k = \sum_{s \leq t} \ell_s^k$ and $L_t^* = \min_k L_t^k$. All losses lie in $[0, 1]$.

Definition 1 (Hedge). For a learning rate $\eta > 0$, Hedge plays $\mathbf{w}_1 = (1/K, \dots, 1/K)$ and

$$w_{t+1}^k(\eta) = \frac{e^{-\eta L_t^k}}{\sum_{k'} e^{-\eta L_t^{k'}}} = \frac{1}{K} \cdot \frac{P_t^k}{B_t}, \quad P_t^k = e^{-\eta L_t^k}, \quad B_t(\eta) = \sum_k \frac{1}{K} e^{-\eta L_t^k}.$$

If we read L_t^k as a negative log-likelihood, $\mathbf{w}_{t+1}$ is the Bayesian posterior distribution over actions using a uniform prior and pseudo-likelihood P_t^k , and B_t is the marginal likelihood. Like a normal marginal likelihood, it breaks down into per-round parts. We use this rule often below.

Lemma 2 (Factorisation). $B_0 = 1$ and $B_t(\eta) = \prod_{s=1}^t (\mathbf{w}_s(\eta) \cdot e^{-\eta \ell_s})$.

Proof. $\frac{B_t}{B_{t-1}} = \sum_k \frac{\frac{1}{K} e^{-\eta L_t^k}}{B_{t-1}} = \sum_k \frac{\frac{1}{K} e^{-\eta L_{t-1}^k}}{B_{t-1}} e^{-\eta \ell_t^k} = \sum_k w_t^k e^{-\eta \ell_t^k} = \mathbf{w}_t \cdot e^{-\eta \ell_t}$, and the claim follows by telescoping. $\square$

Definition 3 (Mixability gap).

$$\delta_t(\eta) = \mathbf{w}_t(\eta) \cdot \ell_t - \left(-\frac{1}{\eta} \ln(\mathbf{w}_t(\eta) \cdot e^{-\eta \ell_t}) \right), \quad \Delta_T(\eta) = \sum_{t=1}^T \delta_t(\eta).$$

The subtracted term is the per-round loss of Vovk's Aggregating Pseudo-Algorithm [4], an ideal strategy that mixes the actions perfectly but cannot be run in practice. So δ_t measures how far the data forces Hedge away from an ideal it cannot reach. The learner can calculate it at the end of round t , which is what lets it be used as a tuning signal.

Proposition 4 (The master decomposition). For every $\eta > 0$ and every T ,

$$\Delta_T(\eta) = \sum_{t=1}^T \mathbf{w}_t \cdot \ell_t + \frac{1}{\eta} \ln B_T(\eta), \quad \text{and} \quad R(T) \leq \frac{\ln K}{\eta} + \Delta_T(\eta). \quad (1)$$

Proof. Summing Definition 3 and applying Lemma 2, $\Delta_T = \sum_t \mathbf{w}_t \cdot \ell_t + \frac{1}{\eta} \sum_t \ln(B_t/B_{t-1}) = \sum_t \mathbf{w}_t \cdot \ell_t + \frac{1}{\eta} \ln B_T$. Dropping all but the best term in $B_T = \sum_k \frac{1}{K} e^{-\eta L_T^k}$ gives $B_T \geq \frac{1}{K} e^{-\eta L_T^*}$, hence $-\frac{1}{\eta} \ln B_T \leq L_T^* + \frac{\ln K}{\eta}$. Therefore $R(T) = \sum_t \mathbf{w}_t \cdot \ell_t - L_T^* = \Delta_T - \frac{1}{\eta} \ln B_T - L_T^* \leq \Delta_T + \frac{\ln K}{\eta}$. $\square$

Equation (1) is the formula the whole paper uses: an exact identity plus one inequality. The only term that depends on the data is $\Delta_T(\eta)$.

Definition 5 (Budget and AdaHedge). The budget is $b(\eta) = (\frac{1}{e-1} + \frac{1}{\eta}) \ln K$. AdaHedge runs Hedge at rate η , and it only monitors Δ during the current segment. It starts a new segment with $\eta \leftarrow \eta/\phi$ as soon as $\Delta \geq b(\eta)$. A restart also resets the weights to uniform, because $\mathbf{w} \propto e^{-\eta \mathbf{L}}$ holds information from the old η and cannot be carried over. Throughout the text, $\phi = 2$, so $\eta_i = \phi^{1-i}$ and $\eta_1 = 1$.

Algorithm 1 AdaHedge(ϕ), $\phi = 2$

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 $\eta \leftarrow \phi$ 
for  $t = 1, 2, \dots$  do
  if  $t = 1$  or  $\Delta \geq b$  then ▷ start a new segment
     $\eta \leftarrow \eta/\phi$ ;  $b \leftarrow (\frac{1}{e-1} + \frac{1}{\eta}) \ln K$ ;  $\Delta \leftarrow 0$ ;  $\mathbf{w} \leftarrow (\frac{1}{K}, \dots, \frac{1}{K})$ 
  end if
  output  $\mathbf{w}$ ; observe  $\ell_t$ 
   $\Delta \leftarrow \Delta + \mathbf{w} \cdot \ell_t + \frac{1}{\eta} \ln(\mathbf{w} \cdot e^{-\eta \ell_t})$ 
   $\mathbf{w} \leftarrow \mathbf{w} \odot e^{-\eta \ell_t} / (\mathbf{w} \cdot e^{-\eta \ell_t})$ 
end for

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Definition 6 (Posterior mass on the leader). $w_t^*(\eta) = \max_{1 \leq k \leq K} w_t^k(\eta)$. Since $w_t^k \propto e^{-\eta L_{t-1}^k}$, this is the probability weight on $\arg \min_k L_{t-1}^k$.

Note — *Three things this report distinguishes carefully.* Confusing the first two is an easy mistake to make here. (i) The **best action** is $\arg \min_k L_T^k$, which is fixed over the whole timeline: it is the baseline inside $R(T)$. (ii) The **current leader** is $\arg \min_k L_t^k$, which is recalculated every round. It is the action carrying w_t^* , and is the focus of Lemmas 15 and 16. It changes often. (iii) In the random data setting of Theorem 19 there is also a **population-best action** k^* , defined by the expected losses. The other two converge to this, but they might not equal it at a specific time t .

3 Proof of a technical result

Our goal is Theorem 19: on independent data with a clear best action, AdaHedge has a regret bounded by a constant that does not grow with T . The proof uses five smaller results. Two of them (Lemmas 8 and 9) are also needed for the worst-case half, so they are proved first and then reused.

Proof map.

result	role
Lemma 7 (elementary inequalities)	the two basic mathematical bounds every later step uses
Lemma 8 ($0 \leq \delta_t \leq \eta/8$)	basic upper bound; also sets the restart limit (14)
Lemma 9 ($\Delta_T \leq (\eta L_T^* + \ln K)/(e-1)$) ↔ Thm. 11, Lemma 12, Thm. 13	makes the budget rule safe the worst-case half
Lemma 14 (log-convexity)	lets Lemma 15 restrict to corner cases
Lemma 15 ($\delta_t \leq (e-2)\eta(1-w_t^*)$)	<i>the improved upper bound</i>
Lemma 16 ($\sum_t (1-w_{t+1}^*) \leq C_K \eta^{-1/\beta}$)	posterior concentration
Lemma 17 (budget never depleted)	⇒ m bounded ⇒ $R = O(1)$
Theorem 19 (stochastic case)	verifies Lemma 17 with high probability

3.1 Elementary tools

Two standard results are used repeatedly and are stated here so the report is self-contained. Both are proved in [2].

Hoeffding's lemma ([2], Lemma A.1): if $X \in [a, b]$ almost surely, then $\ln \mathbb{E}[e^{sX}] \leq s \mathbb{E}[X] + \frac{1}{8} s^2 (b-a)^2$ for every $s \in \mathbb{R}$. *Hoeffding's inequality*: if $X_1, \dots, X_r$ are independent with $X_i \in [0, 1]$ and $S = \sum_i X_i$, then $\Pr(S - \mathbb{E}[S] \geq u) \leq e^{-2u^2/r}$ and $\Pr(S - \mathbb{E}[S] \leq -u) \leq e^{-2u^2/r}$ for every $u > 0$. A third bound, $\ln \mathbb{E}[e^{sX}] \leq (e^s - 1) \mathbb{E}[X]$ for $X \in [0, 1]$, is Lemma A.3 of [2] and is used once, in Lemma 9. We also need two scalar inequalities.

Lemma 7. For all $x \geq 0$, $e^{-x} \leq 1 - x + \frac{1}{2}x^2$. For all $x \in [0, 1]$, $e^x \leq 1 + x + (e-2)x^2$.

Proof. For the first, put $h(x) = 1 - x + \frac{1}{2}x^2 - e^{-x}$. Then $h(0) = 0$, $h'(x) = -1 + x + e^{-x}$ with $h'(0) = 0$, and $h''(x) = 1 - e^{-x} \geq 0$ for $x \geq 0$. So h' is nondecreasing and starts at 0, hence $h' \geq 0$; so h is nondecreasing and starts at 0, hence $h \geq 0$.

For the second, write $g(x) = (e^x - 1 - x)/x^2 = \sum_{j \geq 0} x^j/(j+2)!$ for $x > 0$, extended by $g(0) = \frac{1}{2}$. Every coefficient is positive, so g is nondecreasing on $[0, \infty)$, and $g(1) = e - 2$. Hence $g(x) \leq e - 2$ on $[0, 1]$, which is the claim. $\square$

3.2 The mixability gap: the basic upper bound and the budget rule

Lemma 8 (Paper's Lemma 1). *For every t and every $\eta > 0$, $0 \leq \delta_t(\eta) \leq \eta/8$.*

Proof. Write X for the random variable taking value ℓ_t^k when k is drawn from $\mathbf{w}_t$, so that $\mathbf{w}_t \cdot \ell_t = \mathbb{E}[X]$ and $\mathbf{w}_t \cdot e^{-\eta \ell_t} = \mathbb{E}[e^{-\eta X}]$, and $\delta_t = \mathbb{E}[X] + \frac{1}{\eta} \ln \mathbb{E}[e^{-\eta X}]$.

Lower bound. $\ln$ is a concave function, so Jensen's inequality gives $\ln \mathbb{E}[e^{-\eta X}] \geq \mathbb{E}[\ln e^{-\eta X}] = -\eta \mathbb{E}[X]$, meaning $\delta_t \geq 0$.

Upper bound. $X \in [0, 1]$, so Hoeffding's lemma with $[a, b] = [0, 1]$ gives $\ln \mathbb{E}[e^{sX}] \leq s\mathbb{E}[X] + s^2/8$ for every $s \in \mathbb{R}$. Let $s = -\eta$: $\ln \mathbb{E}[e^{-\eta X}] \leq -\eta \mathbb{E}[X] + \eta^2/8$, and dividing by $\eta > 0$ gives $\delta_t \leq \eta/8$. $\square$

Lemma 8 alone yields $\Delta_T(\eta) \leq \eta T/8$, which combined with (1) gives the standard textbook bound $R(T) \leq \ln K/\eta + \eta T/8$. This is optimised by $\eta = \sqrt{8 \ln K/T}$ at $R(T) \leq \sqrt{T \ln K/2}$. Everything below comes from replacing this simple bound by one that the data can make smaller.

Lemma 9 (Paper's Lemma 2). *For every T and every $\eta \in (0, 1]$, $\Delta_T(\eta) \leq \frac{\eta L_T^* + \ln K}{e - 1}$.*

Proof. For $X \in [0, 1]$ and $s \in \mathbb{R}$ one has $\ln \mathbb{E}[e^{sX}] \leq (e^s - 1)\mathbb{E}[X]$ ([2], Lemma A.3). With X as in the previous proof and $s = -\eta$,

$$\ln(\mathbf{w}_t \cdot e^{-\eta \ell_t}) \leq (e^{-\eta} - 1) \mathbf{w}_t \cdot \ell_t.$$

The factor $e^{-\eta} - 1$ is negative, so dividing by it reverses the inequality: $\frac{1}{e^{-\eta} - 1} \ln(\mathbf{w}_t \cdot e^{-\eta \ell_t}) \geq \mathbf{w}_t \cdot \ell_t$. Substituting this into Definition 3 and writing $f(\eta) = \frac{1}{1 - e^{-\eta}} - \frac{1}{\eta}$,

$$\Delta_T(\eta) = \sum_t \left(\mathbf{w}_t \cdot \ell_t + \frac{1}{\eta} \ln(\mathbf{w}_t \cdot e^{-\eta \ell_t}) \right) \leq \left(\frac{1}{e^{-\eta} - 1} + \frac{1}{\eta} \right) \sum_t \ln(\mathbf{w}_t \cdot e^{-\eta \ell_t}) = -f(\eta) \ln B_T,$$

using Lemma 2 in the last step, and $-\ln B_T \geq 0$.

The last step used two properties of f , and we prove both. Put the two terms over a common denominator:

$$f(\eta) = \frac{1}{1 - e^{-\eta}} - \frac{1}{\eta} = \frac{\eta - (1 - e^{-\eta})}{\eta(1 - e^{-\eta})} = \frac{\eta - 1 + e^{-\eta}}{\eta(1 - e^{-\eta})}. \quad (2)$$

f is nonnegative. In (2) the denominator is positive for $\eta > 0$, because $1 - e^{-\eta} > 0$. The numerator is $\eta - 1 + e^{-\eta}$, which is nonnegative by the first inequality of Lemma 7 read backwards: that inequality gives $e^{-\eta} \geq 1 - \eta$, so $\eta - 1 + e^{-\eta} \geq 0$. Hence $f(\eta) \geq 0$.

f is increasing on $(0, 1]$. Differentiate the original form. Since $\frac{d}{d\eta}(1 - e^{-\eta}) = e^{-\eta}$,

$$f'(\eta) = -\frac{e^{-\eta}}{(1 - e^{-\eta})^2} + \frac{1}{\eta^2}, \quad \text{so} \quad f'(\eta) \geq 0 \iff (1 - e^{-\eta})^2 \geq \eta^2 e^{-\eta}.$$

Both sides are nonnegative, so we may take square roots, which turns the condition into $1 - e^{-\eta} \geq \eta e^{-\eta/2}$. Multiplying through by $e^{\eta/2} > 0$ turns it into

$$e^{\eta/2} - e^{-\eta/2} \geq \eta. \quad (3)$$

Inequality (3) is immediate from the exponential series. Writing $u = \eta/2 \geq 0$ and subtracting the two series term by term, the even powers cancel and the odd ones double:

$$e^u - e^{-u} = \sum_{j \geq 0} \frac{u^j}{j!} - \sum_{j \geq 0} \frac{(-u)^j}{j!} = 2 \sum_{j \text{ odd}} \frac{u^j}{j!} = 2 \left(u + \frac{u^3}{6} + \frac{u^5}{120} + \dots \right) \geq 2u,$$

because every term of the remaining sum is nonnegative for $u \geq 0$. With $u = \eta/2$ this reads $e^{\eta/2} - e^{-\eta/2} \geq \eta$, which is (3). So $f' \geq 0$ and f is increasing.

Therefore $f(\eta) \leq f(1)$ for every $\eta \in (0, 1]$, and $f(1) = \frac{1}{1 - e^{-1}} - 1 = \frac{e}{e - 1} - 1 = \frac{1}{e - 1}$. Finally $B_T \geq \frac{1}{K} e^{-\eta L_T^*}$ gives $-\ln B_T \leq \eta L_T^* + \ln K$, which completes the proof. $\square$

Corollary 10 (Why the budget is the right threshold). *If Hedge is run at $\eta \in (0, 1]$ until the first T with $\Delta_T(\eta) \geq b(\eta)$, then*

$$\eta \geq \sqrt{\frac{(e-1) \ln K}{L_T^*}}. \quad (4)$$

Proof. Combining $b(\eta) \leq \Delta_T(\eta)$ with Lemma 9, $\frac{\ln K}{e-1} + \frac{\ln K}{\eta} \leq \frac{\eta L_T^* + \ln K}{e-1}$. The terms $\frac{\ln K}{e-1}$ cancel, leaving $\frac{\ln K}{\eta} \leq \frac{\eta L_T^*}{e-1}$, i.e. $\eta^2 \geq (e-1) \ln K / L_T^*$. $\square$

Where the budget comes from. The form of $b(\eta)$ is the solution of an optimisation problem, in four steps.

Step 0: what is being minimised. Start from (1), $R(T) \leq \ln K / \eta + \Delta_T(\eta)$. We cannot optimise $\Delta_T(\eta)$ over η , because it is an outcome of the run and not a function known in advance, so we replace it by its bound from Lemma 9. Everything is then an explicit function of η and L_T^* :

$$R(T) \leq F(\eta) := \underbrace{\frac{\ln K}{\eta}}_{\text{decreasing}} + \underbrace{\frac{\eta L_T^*}{e-1}}_{\text{increasing}} + \underbrace{\frac{\ln K}{e-1}}_{\text{constant in } \eta}.$$

Step 1: minimise F . Write $A = \ln K$ and $c = L_T^* / (e-1)$. Then

$$F'(\eta) = -\frac{A}{\eta^2} + c = 0 \implies \eta_{\text{opt}} = \sqrt{\frac{A}{c}} = \sqrt{\frac{(e-1) \ln K}{L_T^*}}, \quad F''(\eta) = \frac{2A}{\eta^3} > 0,$$

so this is a minimum. At that point the two η -dependent terms are equal: $A / \eta_{\text{opt}} = \sqrt{Ac} = c \eta_{\text{opt}}$, that is $\ln K / \eta_{\text{opt}} = \eta_{\text{opt}} L_T^* / (e-1)$.

Step 2: invert. η_{opt} is a function of L_T^* . Inverting it answers the opposite question: for a *given* η , at which horizon is η exactly optimal?

$$L^*(\eta) = \frac{(e-1) \ln K}{\eta^2}.$$

Step 3: define b . Take the budget to be the Lemma 9 ceiling evaluated at that point:

$$b(\eta) := \frac{\eta L^*(\eta) + \ln K}{e-1} = \frac{1}{e-1} \left(\frac{(e-1) \ln K}{\eta} + \ln K \right) = \frac{\ln K}{\eta} + \frac{\ln K}{e-1},$$

which is Definition 5.

Step 4: what the test means. $b(\eta)$ is the largest Δ consistent with the hypothesis that η is still the right rate, so the test $\Delta \geq b(\eta)$ rejects that hypothesis: it is a statement about Δ , not a bound on the regret. Read in the other direction it is Corollary 10, since

$$\frac{\ln K}{\eta} + \frac{\ln K}{e-1} \leq \Delta_T \leq \frac{\eta L_T^*}{e-1} + \frac{\ln K}{e-1} \implies \frac{\ln K}{\eta} \leq \frac{\eta L_T^*}{e-1} \implies \eta \geq \sqrt{\frac{(e-1) \ln K}{L_T^*}},$$

the term $\ln K / (e-1)$ cancelling on both sides. Evaluating F at its minimum gives $F(\eta_{\text{opt}}) = \sqrt{\frac{4}{e-1} L_T^* \ln K} + \frac{\ln K}{e-1}$, which is exactly the bound of Theorem 11 below, apart from the $1/8$ that comes from $\delta_t \leq \eta/8$ at the crossing round.

Theorem 11 (Paper's Theorem 3). *Suppose Hedge runs at $\eta \in (0, 1]$ and after T rounds it has just used up the budget, meaning $b(\eta) \leq \Delta_T(\eta) < b(\eta) + \eta/8$. Then $R(T) < \sqrt{\frac{4}{e-1} L_T^* \ln K} + \frac{1}{e-1} \ln K + \frac{1}{8}$.*

Proof. Using the formula in (1) and $B_T \geq \frac{1}{K} e^{-\eta L_T^*}$,

$$\sum_{t=1}^T \mathbf{w}_t \cdot \ell_t = \Delta_T(\eta) - \frac{1}{\eta} \ln B_T < b(\eta) + \frac{\eta}{8} + \frac{1}{\eta} (\eta L_T^* + \ln K) \leq \frac{\ln K}{e-1} + \frac{1}{8} + \frac{2 \ln K}{\eta} + L_T^*, \quad (5)$$

where $\eta \leq 1$ was used for $\eta/8 \leq 1/8$ and $b(\eta)$ contributed one of the two $\ln K / \eta$ terms. Subtracting L_T^* and inserting (4) into $2 \ln K / \eta \leq 2 \ln K \sqrt{L_T^* / ((e-1) \ln K)} = \sqrt{4 L_T^* \ln K / (e-1)}$ completes the proof. $\square$

Lemma 12 (Paper's Lemma 4). *If AdaHedge has started m segments after T rounds, then*

$$R(T) < 2 \ln K \frac{\phi^m - 1}{\phi - 1} + m \left(\frac{\ln K}{e - 1} + \frac{1}{8} \right).$$

Proof. Bound (5) applies in exactly the same way to each segment, with L_T^* replaced by $L^*(i)$, the loss of the best action on segment i . It also works for an incomplete final segment, because there $\Delta < b(\eta)$ and the inequality is only made looser.

Summing the m per-segment bounds, we need $\sum_{i=1}^m L^*(i) \leq L_T^*$. This holds because $L^*(i) = \min_k L^k(i) \leq L^{k^*}(i)$ for the globally best action k^* , and these segment losses add up to $\sum_{i=1}^m L^{k^*}(i) = L_T^{k^*} = L_T^*$. The rates satisfy $\sum_{i=1}^m 1/\eta_i = \sum_{i=0}^{m-1} \phi^i = (\phi^m - 1)/(\phi - 1)$. $\square$

Theorem 13 (Paper's Theorem 5 — the worst-case half). $R(T) \leq \frac{\phi \sqrt{\phi^2 - 1}}{\phi - 1} \sqrt{\frac{4}{e - 1} L_T^* \ln K} + O(\ln(L_T^* + 2) \ln K)$, with leading factor 3.46 at $\phi = 2$.

Proof. Write $L^*(i)$ for the loss of the best action on segment i alone, that is $L^*(i) = \min_k L^k(i)$ where $L^k(i)$ is the loss action k accumulates during segment i .

Step 1: bound the number of segments. If $m = 1$ the theorem follows from Lemma 12 directly, so assume $m \geq 2$. Then segments $1, \dots, m - 1$ each ran until their budget was used up, so Corollary 10 applies to each of them separately, with L_T^* replaced by that segment's own $L^*(i)$:

$$\eta_i \geq \sqrt{\frac{(e - 1) \ln K}{L^*(i)}}, \quad \text{equivalently} \quad \frac{1}{\eta_i^2} \leq \frac{L^*(i)}{(e - 1) \ln K}, \quad i = 1, \dots, m - 1.$$

Sum these over i . On the left, $\eta_i = \phi^{1-i}$ gives $1/\eta_i^2 = \phi^{2i-2}$, a geometric sum:

$$\sum_{i=1}^{m-1} \frac{1}{\eta_i^2} = \sum_{i=1}^{m-1} \phi^{2i-2} = \sum_{i=0}^{m-2} (\phi^2)^i = \frac{\phi^{2m-2} - 1}{\phi^2 - 1}.$$

On the right, $\sum_{i=1}^{m-1} L^*(i) \leq L_T^*$ by the argument already used in Lemma 12: for the globally best action k^* we have $L^*(i) \leq L^{k^*}(i)$ for each i , and those segment losses add up to $L_T^{k^*} = L_T^*$. Combining,

$$\frac{\phi^{2m-2} - 1}{\phi^2 - 1} \leq \frac{L_T^*}{(e - 1) \ln K}, \quad \text{so} \quad \phi^{2m-2} \leq A + 1, \quad A := \frac{(\phi^2 - 1) L_T^*}{(e - 1) \ln K}. \quad (6)$$

Taking $\log_\phi$ of (6) and dividing by 2 gives $m \leq 1 + \frac{1}{2} \log_\phi(A + 1)$, so $m = O(\ln(L_T^* + 2))$.

Step 2: substitute into Lemma 12. That lemma states

$$R(T) < \underbrace{\frac{2 \ln K (\phi^m - 1)}{\phi - 1}}_{\text{(I)}} + \underbrace{m \left(\frac{\ln K}{e - 1} + \frac{1}{8} \right)}_{\text{(II)}}.$$

Term (II) is $O(\ln(L_T^* + 2) \ln K)$ by Step 1, so only (I) needs care. Write $\phi^m = \phi \cdot \phi^{m-1} = \phi \sqrt{\phi^{2m-2}}$ and insert (6):

$$\phi^m \leq \phi \sqrt{A + 1} \leq \phi(\sqrt{A} + 1),$$

the second step by $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$ with $a = A$ and $b = 1$. Hence

$$\text{(I)} \leq \frac{2 \ln K}{\phi - 1} (\phi \sqrt{A} + \phi - 1) = \frac{2 \phi \ln K}{\phi - 1} \sqrt{A} + 2 \ln K.$$

Step 3: simplify the leading term. Move the factor $\ln K$ inside the square root:

$$\ln K \cdot \sqrt{A} = \sqrt{(\ln K)^2 \cdot \frac{(\phi^2 - 1) L_T^*}{(e - 1) \ln K}} = \sqrt{\frac{(\phi^2 - 1) L_T^* \ln K}{e - 1}} = \sqrt{\phi^2 - 1} \sqrt{\frac{L_T^* \ln K}{e - 1}}.$$

Therefore

$$\frac{2 \phi \ln K}{\phi - 1} \sqrt{A} = \frac{2 \phi \sqrt{\phi^2 - 1}}{\phi - 1} \sqrt{\frac{L_T^* \ln K}{e - 1}} = \frac{\phi \sqrt{\phi^2 - 1}}{\phi - 1} \sqrt{\frac{4}{e - 1} L_T^* \ln K},$$

where the last step used $2\sqrt{z} = \sqrt{4z}$. Collecting this with the $2\ln K$ from Step 2 and with term (II) gives

$$R(T) \leq \frac{\phi\sqrt{\phi^2-1}}{\phi-1} \sqrt{\frac{4}{e-1} L_T^* \ln K} + O(\ln(L_T^* + 2) \ln K),$$

which is the statement.

Step 4: the constant. The leading factor is $h(\phi) = \phi\sqrt{\phi^2-1}/(\phi-1)$. At $\phi = 2$ this is $2\sqrt{3}/1 = 3.464$. Minimising h over $\phi > 1$ gives the golden ratio $\phi = (1 + \sqrt{5})/2$, where $h = 3.330$, so the simpler choice $\phi = 2$ costs about 4% in the leading constant. $\square$

Everything so far is worst-case: the only detail we used about the data was L_T^* . Now we start the second part.

3.3 The improved upper bound: convexity and the posterior

Lemma 14 (Paper's Lemma 10 — log-convexity). *For every $\eta > 0$ and every $\mathbf{w}$ in the simplex, the function $\ell \mapsto \ln(\mathbf{w} \cdot e^{-\eta\ell})$ is convex on $\mathbb{R}^K$.*

The proof uses Hölder's inequality, so we state it first in the form we need.

Hölder's inequality. Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, and let X, Y be random variables. Then

$$\mathbb{E}[|XY|] \leq \mathbb{E}[|X|^p]^{1/p} \mathbb{E}[|Y|^q]^{1/q}. \quad (7)$$

Here is why we need it. Convexity of $\ln(\mathbf{w} \cdot e^{-\eta\ell})$ is a statement about a *mixture*. We know how each single term $e^{-\eta\ell^k}$ behaves between two loss vectors, and we need that behaviour to survive after averaging over k with weights $\mathbf{w}$.

Each term satisfies a product identity, one k at a time. Hölder's inequality (7) is the tool that turns a bound on the expectation of a *product* into a product of two separate expectations. We choose the exponents $p = 1/(1-\lambda)$ and $q = 1/\lambda$ because they make the powers cancel, and what is left is exactly the two quantities the definition of convexity asks for.

Proof of Lemma 14. Fix loss vectors ℓ_0, ℓ_1 and $\lambda \in (0, 1)$, and put $\ell_\lambda = (1-\lambda)\ell_0 + \lambda\ell_1$. Let $g_k(\ell) = e^{-\eta\ell^k}$.

Step 1: each term is log-affine. Since $\ell_\lambda^k = (1-\lambda)\ell_0^k + \lambda\ell_1^k$,

$$\ln g_k(\ell_\lambda) = -\eta \ell_\lambda^k = (1-\lambda)(-\eta \ell_0^k) + \lambda(-\eta \ell_1^k) = (1-\lambda) \ln g_k(\ell_0) + \lambda \ln g_k(\ell_1),$$

and exponentiating gives the pointwise identity

$$g_k(\ell_\lambda) = g_k(\ell_0)^{1-\lambda} g_k(\ell_1)^\lambda \quad \text{for every } k. \quad (8)$$

Step 2: average over k and apply Hölder. Take expectations of (8) with $k \sim \mathbf{w}$. In (7) set

$$X = g_k(\ell_0)^{1-\lambda}, \quad Y = g_k(\ell_1)^\lambda, \quad p = \frac{1}{1-\lambda}, \quad q = \frac{1}{\lambda}.$$

These exponents are admissible: both exceed 1 because $\lambda \in (0, 1)$, and $\frac{1}{p} + \frac{1}{q} = (1-\lambda) + \lambda = 1$. They were chosen so that the powers cancel, $|X|^p = g_k(\ell_0)$ and $|Y|^q = g_k(\ell_1)$, and both variables are positive so the absolute values can be dropped. Substituting into (7),

$$\mathbb{E}_{k \sim \mathbf{w}}[g_k(\ell_\lambda)] = \mathbb{E}_{k \sim \mathbf{w}}[g_k(\ell_0)^{1-\lambda} g_k(\ell_1)^\lambda] \leq \mathbb{E}_{k \sim \mathbf{w}}[g_k(\ell_0)]^{1-\lambda} \mathbb{E}_{k \sim \mathbf{w}}[g_k(\ell_1)]^\lambda.$$

Step 3: take logarithms. Each expectation $\mathbb{E}_{k \sim \mathbf{w}}[g_k(\ell)]$ is exactly $\mathbf{w} \cdot e^{-\eta\ell}$. Taking logarithms of both sides, and using $\ln(a^{1-\lambda}b^\lambda) = (1-\lambda)\ln a + \lambda\ln b$,

$$\ln(\mathbf{w} \cdot e^{-\eta\ell_\lambda}) \leq (1-\lambda)\ln(\mathbf{w} \cdot e^{-\eta\ell_0}) + \lambda\ln(\mathbf{w} \cdot e^{-\eta\ell_1}),$$

which is the definition of convexity. $\square$

Lemma 15 (Paper's Lemma 6 — the improved upper bound). *For every t and every $\eta \in (0, 1]$, $\delta_t(\eta) \leq (e-2)\eta(1-w_t^*(\eta))$.*

Proof. Fix t and write $\mathbf{w} = \mathbf{w}_t(\eta)$. As a function of ℓ_t , the mixability gap $\delta_t = \mathbf{w} \cdot \ell_t + \frac{1}{\eta} \ln(\mathbf{w} \cdot e^{-\eta \ell_t})$ is the sum of a linear term and, by Lemma 14, a convex term, so it is convex.

A convex function on a compact convex set attains its maximum at an extreme point of that set. The box $[0, 1]^K$ is compact and convex, and its extreme points are exactly the vectors whose every coordinate is 0 or 1: any point with some coordinate strictly inside $(0, 1)$ is the midpoint of two other points of the box, so it is not extreme. Hence it is enough to prove the bound when every $\ell_t^k \in \{0, 1\}$.

Assume this, and let $p = \mathbf{w} \cdot \ell_t$ be the total weight that the round- t distribution places on the actions whose loss is 1. (The letter α is reserved for the gap constant of Lemmas 16 and 17, so it is not reused here.) Then $\mathbf{w} \cdot e^{-\eta \ell_t} = (1 - p) + pe^{-\eta}$ and

$$\delta_t = p + \frac{1}{\eta} \ln(1 + p(e^{-\eta} - 1)). \quad (9)$$

We bound this in two ways.

p near 0. Using $\ln x \leq x - 1$ and then $e^{-\eta} \leq 1 - \eta + \frac{1}{2}\eta^2$ (Lemma 7),

$$\delta_t \leq p + \frac{1}{\eta} p(e^{-\eta} - 1) \leq p + \frac{p}{\eta}(-\eta + \frac{1}{2}\eta^2) = \frac{1}{2}p\eta.$$

p near 1. Rewrite (9) by extracting e^η inside the logarithm: $\frac{1}{\eta} \ln(1 + p(e^{-\eta} - 1)) = -1 + \frac{1}{\eta} \ln(e^\eta(1 - p) + p)$, so $\delta_t = p - 1 + \frac{1}{\eta} \ln(e^\eta(1 - p) + p)$. Now $\ln x \leq x - 1$ gives $\ln(e^\eta(1 - p) + p) \leq (1 - p)(e^\eta - 1)$, and $e^\eta - 1 \leq \eta + (e - 2)\eta^2$ for $\eta \leq 1$ (Lemma 7). Hence

$$\delta_t \leq p - 1 + \frac{1 - p}{\eta}(\eta + (e - 2)\eta^2) = p - 1 + (1 - p) + (e - 2)\eta(1 - p) = (e - 2)\eta(1 - p).$$

Since $\frac{1}{2} \leq e - 2$, the two bounds combine into $\delta_t \leq (e - 2)\eta \min\{p, 1 - p\}$.

It remains to relate $\min\{p, 1 - p\}$ to w_t^* . Recall from Definition 6 that $w_t^*(\eta) = \max_{1 \leq k \leq K} w_t^k(\eta)$, the largest weight the round- t distribution puts on any single action. Let $k^\dagger$ be an action attaining it, with $w^{k^\dagger} = w_t^*$. If $\ell_t^{k^\dagger} = 0$ then $k^\dagger$ does not contribute to p , so $p \leq 1 - w_t^*$. If instead $\ell_t^{k^\dagger} = 1$ then $k^\dagger$ does contribute, so $p \geq w_t^*$ and therefore $1 - p \leq 1 - w_t^*$. In both cases $\min\{p, 1 - p\} \leq 1 - w_t^*$. $\square$

This is the most important point of the whole paper. Lemma 8 bounds δ_t using a value that depends only on η . Lemma 15 bounds it using a value that the data can push down to zero. If the probability weight concentrates on one action, the mixability gap stops growing.

3.4 From posterior concentration to a bounded number of segments

Lemma 16 (Paper's Lemma 7). *Let $\alpha, \beta > 0$ and $\tau \in \mathbb{Z}^+$. Suppose that for times $t = \tau, \tau + 1, \dots, T$ one single action k^* stays in the lead with $L_t^{k^*} = L_t^*$. Suppose also that the other cumulative losses move away from it, so that $L_t^k - L_t^* \geq \alpha t^\beta$ for all $k \neq k^*$. Then for every $\eta > 0$*

$$\sum_{t=\tau}^T (1 - w_{t+1}^*(\eta)) \leq C_K \eta^{-1/\beta}, \quad C_K = (K - 1) \alpha^{-1/\beta} \Gamma\left(1 + \frac{1}{\beta}\right),$$

which is a constant that does not depend on η, τ or T .

Proof. Since k^* is the only leader at these times, $w_{t+1}^* = w_{t+1}^{k^*}$, and by Definition 1

$$\frac{1}{w_{t+1}^*(\eta)} = \sum_{k=1}^K e^{-\eta(L_t^k - L_t^*)} \leq 1 + (K - 1)e^{-\alpha \eta t^\beta},$$

because the $k = k^*$ term adds 1 and each other term is at most $e^{-\alpha \eta t^\beta}$. Writing $u = (K - 1)e^{-\alpha \eta t^\beta}$, we get $w_{t+1}^* \geq 1/(1 + u)$, so $1 - w_{t+1}^* \leq u/(1 + u) \leq u$. Therefore

$$\sum_{t=\tau}^T (1 - w_{t+1}^*) \leq (K - 1) \sum_{t=\tau}^T e^{-\alpha \eta t^\beta} \leq (K - 1) \int_0^\infty e^{-\alpha \eta t^\beta} dt,$$

the last step works because the function is nonincreasing in t and $\tau \geq 1$. Substituting $u = \alpha \eta t^\beta$, so $t = (u/\alpha \eta)^{1/\beta}$ and $dt = \frac{1}{\beta}(\alpha \eta)^{-1/\beta} u^{1/\beta - 1} du$,

$$\int_0^\infty e^{-\alpha \eta t^\beta} dt = \frac{1}{\beta}(\alpha \eta)^{-1/\beta} \int_0^\infty e^{-u} u^{1/\beta - 1} du = \frac{1}{\beta}(\alpha \eta)^{-1/\beta} \Gamma\left(\frac{1}{\beta}\right) = (\alpha \eta)^{-1/\beta} \Gamma\left(1 + \frac{1}{\beta}\right),$$

using the rule $z\Gamma(z) = \Gamma(1 + z)$ with $z = 1/\beta$. $\square$

Lemma 17 (Paper's Lemma 8). *Let $\alpha > 0$, $\beta > 1/2$, and let C_K be as in Lemma 16. Suppose AdaHedge starts a segment m^* such that*

$$\tau := \lceil 8 \ln(K) \phi^{(m^*-1)(2-1/\beta)} - 8(e-2)C_K + 1 \rceil \geq 1, \quad (10)$$

and such that, writing $L_r^k(m^)$ for the loss of action k over the first r rounds of that segment, some action k^* satisfies $L_r^k(m^*) - L_r^{k^*}(m^*) \geq \alpha r^\beta$ for all $r \geq \tau$ and all $k \neq k^*$. Then AdaHedge starts at most m^* segments, and consequently $R(T) = O(1)$.*

Proof. Let $\eta = \eta_{m^*} = \phi^{1-m^*} \leq 1$ and let $\Delta_T(m^*, \eta)$ be the cumulative mixability gap measured only from the start of segment m^* . We split the sum at τ : we bound the first $\tau - 1$ terms using Lemma 8 and the rest using Lemma 15 followed by Lemma 16,

$$\Delta_T(m^*, \eta) \leq \frac{(\tau-1)\eta}{8} + (e-2)\eta \cdot C_K \eta^{-1/\beta} \leq \frac{1}{8}(\tau-1 + 8(e-2)C_K)\eta^{1-1/\beta},$$

where the last step works because $\eta \leq \eta^{1-1/\beta}$ is valid when $\eta \leq 1$.

The budget for segment m^* is never used up as long as $\Delta_T(m^*, \eta) < b(\eta)$, and $\frac{1}{8}(\tau-1+8(e-2)C_K)\eta^{1-1/\beta} \leq \ln K/\eta$ is enough to guarantee this. Rearranging the terms gives

$$\tau \leq 8 \ln(K) \eta^{1/\beta-2} - 8(e-2)C_K + 1 = 8 \ln(K) \phi^{(m^*-1)(2-1/\beta)} - 8(e-2)C_K + 1,$$

which is exactly (10). Therefore no new segment is started, $m \leq m^*$, and Lemma 12 bounds $R(T)$ by a value that depends on m^* but not on T . $\square$

Remark 18 (Where $\beta > 1/2$ is used). The exponent of ϕ in (10) is $(m^*-1)(2-1/\beta)$, which is only positive when $\beta > 1/2$. If $\beta \leq 1/2$, the right-hand side does not grow with m^* , so no segment would satisfy the condition and the lemma would be useless. Separation that happens more slowly than $\sqrt{t}$ is outside the scope of this argument. And $\sqrt{t}$ is exactly the scale of the normal random fluctuations when summing t independent bounded losses, so this limit makes statistical sense.

3.5 The stochastic case

Theorem 19 (Paper's Theorem 9). *Let $\alpha > 0$ and $\delta \in (0, 1]$, and let k^* be a fixed action. Suppose the loss vectors ℓ_t are independent random variables whose expected differences satisfy*

$$\min_{k \neq k^*} \mathbb{E}[\ell_t^k - \ell_t^{k^*}] \geq 2\alpha \quad \text{for all } t \in \mathbb{Z}^+. \quad (11)$$

Then, with probability at least $1 - \delta$, AdaHedge starts at most

$$m^* = 1 + \left\lceil \log_\phi \left(\frac{(K-1)(e-2)}{\alpha \ln K} + \frac{\ln(2K/(\alpha^2 \delta))}{4\alpha^2 \ln K} + \frac{1}{8 \ln K} \right) \right\rceil \quad (12)$$

segments, and consequently $R(T) = O(K + \log(1/\delta))$.

Proof. We check the requirements of Lemma 17 using $\beta = 1$ and this α , with probability at least $1 - \delta$. With $\beta = 1$, Lemma 16 gives $C_K = (K-1)\alpha^{-1}\Gamma(2) = (K-1)/\alpha$.

Step 1: defining a sufficient event. For $r = \tau, \tau + 1, \dots$ let

$$A_r^k = \left\{ L_r^k(m^*) - \mathbb{E}[L_r^k(m^*)] \geq -\frac{\alpha}{2}r \right\} \quad (k \neq k^*), \quad B_r = \left\{ L_r^{k^*}(m^*) - \mathbb{E}[L_r^{k^*}(m^*)] \leq \frac{\alpha}{2}r \right\},$$

and $D_r = B_r \cap \bigcap_{k \neq k^*} A_r^k$. On the event D_r , using (11) which means $\mathbb{E}[L_r^k - L_r^{k^*}] \geq 2\alpha r$, we get

$$L_r^k - L_r^{k^*} \geq \mathbb{E}[L_r^k] - \frac{\alpha}{2}r - \mathbb{E}[L_r^{k^*}] - \frac{\alpha}{2}r \geq 2\alpha r - \alpha r = \alpha r,$$

which is exactly the separation condition needed for Lemma 17 when $\beta = 1$. So we just need $\bigcap_{r \geq \tau} D_r$ to hold.

Step 2: calculating the probability. Each $L_r^k(m^*)$ is a sum of r independent variables in $[0, 1]$, so Hoeffding's inequality with $u = \frac{\alpha}{2}r$ gives $\Pr(\bar{A}_r^k) \leq \exp(-2(\frac{\alpha}{2}r)^2/r) = \exp(-\frac{1}{2}\alpha^2 r)$, and the same for $\Pr(\bar{B}_r)$. A union bound over the K events gives $\Pr(\bar{D}_r) \leq K \exp(-\frac{1}{2}\alpha^2 r)$, and a second union bound over r gives

$$\Pr\left(\bigcup_{r \geq \tau} \bar{D}_r\right) \leq \sum_{r=\tau}^{\infty} K e^{-\frac{1}{2}\alpha^2 r} \leq K \int_{\tau-1}^{\infty} e^{-\frac{1}{2}\alpha^2 r} dr = \frac{2K}{\alpha^2} e^{-\frac{1}{2}\alpha^2(\tau-1)}.$$

For this probability to be at most δ , we need

$$\tau \geq \frac{2}{\alpha^2} \ln\left(\frac{2K}{\alpha^2\delta}\right) + 1. \quad (13)$$

Step 3: matching the two definitions of τ . With $\beta = 1$, definition (10) reads $\tau = \lfloor 8 \ln(K) \phi^{m^*-1} - 8(e-2)(K-1)/\alpha + 1 \rfloor$. So (13) is satisfied if

$$8 \ln(K) \phi^{m^*-1} - \frac{8(K-1)(e-2)}{\alpha} \geq \frac{2}{\alpha^2} \ln\left(\frac{2K}{\alpha^2\delta}\right) + 1.$$

We now solve this for m^* . Divide both sides by $8 \ln K$:

$$\phi^{m^*-1} \geq \frac{(K-1)(e-2)}{\alpha \ln K} + \frac{\ln(2K/(\alpha^2\delta))}{4\alpha^2 \ln K} + \frac{1}{8 \ln K} =: X,$$

where the three terms come from $\frac{8(K-1)(e-2)}{\alpha}$, $\frac{2}{\alpha^2} \ln(\cdot)$ and 1 respectively, each divided by $8 \ln K$. Taking $\log_\phi$ gives $m^* - 1 \geq \log_\phi X$, and since m^* must be an integer it is enough to take $m^* = 1 + \lceil \log_\phi X \rceil$, which is exactly (12).

Step 4: proving $\tau \geq 1$. Condition (11) forces $2\alpha \leq 1$, because losses must lie in $[0, 1]$. This means $\frac{2}{\alpha^2} \ln(2K/(\alpha^2\delta)) \geq 0$, so (13) already guarantees $\tau \geq 1$.

Step 5: the size of the bound. Lemma 17 applies, so the algorithm starts at most m^* segments, and Lemma 12 turns this into a bound on $R(T)$ that does not contain T . To see the size of that bound, write X for the bracket inside (12). Then $m^* \leq 2 + \log_\phi X$, so $\phi^{m^*} \leq \phi^2 X$, and the leading term of Lemma 12 is at most

$$\frac{2 \ln K \phi^2}{\phi - 1} X = \frac{2\phi^2}{\phi - 1} \left(\frac{(K-1)(e-2)}{\alpha} + \frac{\ln(2K/(\alpha^2\delta))}{4\alpha^2} + \frac{1}{8} \right).$$

For a fixed α the first term is $O(K)$ and the second is $O(\log(1/\delta))$, so the whole bound is $O(K + \log(1/\delta))$, as claimed. Section 4 evaluates this expression numerically, where the hidden dependence on α turns out to matter a great deal. $\square$

Remark 20 (What the theorem does and does not mean). It does not mean that the regret stops growing after some specific round. It means that the regret is bounded by a number that does not depend on T . It also does not promise this for every single run: the failure probability δ covers the whole timeline at once, which is why $\log(1/\delta)$ appears. The size of that constant is what decides whether the theorem says anything useful at a given horizon.

4 Some critical considerations

4.1 The AP dataset, and why it was built

The problem gives us two extreme settings and nothing in between. In **stochastic** one action is Bern(0.3) and three are Bern(0.5): a single action is best the whole time, the probability weight concentrates on it, and the requirement of Lemma 16 is met perfectly. In **adversarial** the pattern is fixed, action $t \bmod K$ takes loss 1 and the others 0: no action is ever better, the weight never leaves $1/K$, and the requirement fails completely. A third setting, **low-gap**, is **stochastic** with the gap shrunk from 0.20 to 0.01.

Theorem 19 is a chain of four steps, and two data points cannot show whether a chain behaves like a chain. **AP** is our attempt to fill the space between them. It is an artificial dataset with a single dial, the number of windows n , and turning that dial moves the problem from one end of the range to the other.

We make no claim that it is a deep theory of what makes data hard. It is the simplest construction we could write down that turns “easy” and “hard” into a range we can walk along, measuring at every step.

Definition 21 (AP — adversarial permutation). Fix the number of windows n dividing T , so each window has width $W = T/n$. Inside window i :

- (i) every action gets $\lfloor W/2 \rfloor$ losses of one, placed in a random order in its own column, so each action starts at 50% of the window;
- (ii) a *window leader* is chosen at random among the $K - 1$ actions that did not lead window $i - 1$;
- (iii) x_i of the leader’s ones are removed and *discarded*, where x_i ramps linearly so that the leader ends the last window at 30% of W .

When the windows are too short for 30% to be a whole number of losses, the ramp is taken all the way down instead, to 0%. We use $n \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000\}$.

Few windows are wide, and a wide window gives one action a long, clearly visible advantage. Many windows are short, and short windows give many small advantages that cancel. At $n = 1$ there is a single window, one action is favoured for the whole horizon, and the construction should look like **stochastic**; at $n = 5000$ a window is two rounds long and can barely hold an advantage at all.

Table 1: The AP family, indexed by the number of windows n . The leader ends at 30% of the window whenever $W \geq 10$. At $n = 2000$ the windows are five rounds long and the counts are too coarse for 50% and 30% ($\lfloor 5/2 \rfloor = 2$ and $\lfloor 0.3 \cdot 5 \rfloor = 1$); at $n = 5000$ the ramp falls through to 0%.

n	$W = T/n$	start	end	$x_1 \rightarrow x_n$	ones discarded	total ones
1	10 000	50%	30%	2000 $\rightarrow$ 2000	2000	18 000
2	5000	50%	30%	500 $\rightarrow$ 1000	1500	18 500
5	2000	50%	30%	80 $\rightarrow$ 400	1200	18 800
10	1000	50%	30%	20 $\rightarrow$ 200	1100	18 900
20	500	50%	30%	5 $\rightarrow$ 100	1050	18 950
50	200	50%	30%	0 $\rightarrow$ 40	1000	19 000
100	100	50%	30%	0 $\rightarrow$ 20	970	19 030
200	50	50%	30%	0 $\rightarrow$ 10	910	19 090
500	20	50%	30%	0 $\rightarrow$ 4	754	19 246
1000	10	50%	30%	0 $\rightarrow$ 2	502	19 498
2000	5	40%	20%	0 $\rightarrow$ 1	4001	15 999
5000	2	50%	0%	0 $\rightarrow$ 1	1	19 999

Remark 22 (The window leader is not the overall leader). Step (ii) decides who is favoured *inside* a window. Which action is winning overall is a different question. Write κ_j for the leader of window j (a different object from the loss vector ℓ_j), and let $A_k(i) = \sum_{j \leq i: \kappa_j = k} x_j$ be the total advantage action k has collected. Every window subtracts x_j from its own leader and nothing from anyone else. So, up to the ordering inside the window, $L_{t(i)}^k = i \lfloor W/2 \rfloor - A_k(i)$, and therefore $\arg \min_k L^k$ is $\arg \max_k A_k(i)$. That is the action with the largest *total* advantage, not the one leading now.

Measured at window ends, the window leader is also the action of smallest overall loss 100% of the time at $n = 1$ (there is only one window), falling to 78.4% at $n = 10$, 43.0% at $n = 100$, 27.2% at $n = 1000$ and 25.1% at $n = 5000$. The last figure is chance for $K = 4$: by then the windows are too short for any advantage to survive, and the leader label no longer tracks anything in the data.

4.2 Testing the chain, step by step

All measurements use $T = 10^4$, $K = 4$, losses in $\{0, 1\}$, and 50 seeds fixed in advance and shared by every setting. Hedge appears at $2\eta^*$, η^* and $\eta^*/2$, where $\eta^* = \sqrt{2 \ln K / L_T^*}$ is read off each loss matrix after the run. That gives Hedge knowledge AdaHedge never has, so every comparison below is unfair on purpose in Hedge’s favour.

The twelve AP rows from $n = 1$ up to $n = 5000$ form one smooth sequence. Reading down, w_T^* falls from 1.000 to 0.254 and $\Delta_T(1)$ rises from 1.77 to 892, both always moving in the same direction as n grows and with no jumps. The two ends land on the two given settings: AP-1 sits on top of **stochastic** (1.000 against 1.000, 1.77 against 1.54) and AP-5000 on top of **adversarial** (0.254 against 0.246, 892 against 970). One dial, the number of windows, moves the problem all the way.

Two rows deserve a note. In **stochastic**, $\Delta_T(1) = 1.54$ against a budget of 2.193: the budget is never used up, AdaHedge stays on one segment at $\eta = 1$ for the whole run ($m = 1$ on 42 of 50 runs), and its regret curve is flat after a few hundred rounds. That is the mechanism of Lemma 17 seen directly, and AP-1 reproduces it. In **adversarial**, $\Delta_T(1) = 969.55$ against the Lemma 8 bound of 1250: the data produce 78% of the largest gap any sequence can, and AP-5000 reaches 71%.

Note — *Why m stops at six.* The segment count stops at 6 in the lower half of the table. The limit comes from the horizon, not from the data. Equation (14) below gives $T_{\min}(\eta) = b(\eta)/(\eta/8)$, the number of rounds a segment must take whatever the losses are. Adding these up, the seventh segment cannot start before round 15 545, which is past T . So $m \leq 6$ is forced here, and the m column is a real measurement only in the upper rows. This is why Table 2 is ordered by $\Delta_T(1)$ and not by m .

4.3 What really makes the data difficult

It is easy to read Table 2 as saying the difficulty comes from changing the leader more often. It does not, and the sweep itself contains the control.

Table 2: Every step of $w^* \rightarrow \Delta_T \rightarrow \text{budget} \rightarrow m \rightarrow R$, over 50 runs, ordered by $\Delta_T(1)$; mean $\pm$ sd. w_T^* is the weight on the overall best action at $t = T$ under Hedge($2\eta^*$), and $\Delta_T(1)$ is the cumulative mixability gap of Hedge at $\eta = 1$, to be compared with $b(1) = 2.193$ and with the Lemma 8 bound $\eta T/8 = 1250$. The three non-AP settings are marked in italics.

regime	w_T^*	$\Delta_T(1)$	$b(1)$ used up?	m	R_{AdaHedge}
<i>stochastic</i>	1.000 ± 0.000	1.54 ± 0.96	no	1.18 ± 0.43	3.5 ± 2.3
AP-1	1.000 ± 0.000	1.77 ± 1.00	no	1.30 ± 0.50	4.1 ± 2.8
AP-2	1.000 ± 0.000	5.18 ± 2.95	yes	2.46 ± 0.75	8.1 ± 5.6
AP-5	0.895 ± 0.199	17.57 ± 6.18	yes	4.34 ± 0.59	-23.9 ± 69.4
AP-10	0.878 ± 0.167	39.50 ± 9.67	yes	5.00 ± 0.00	36.1 ± 23.6
<i>low-gap</i>	0.863 ± 0.146	43.22 ± 24.37	yes	5.46 ± 0.78	59.7 ± 22.8
AP-20	0.836 ± 0.169	69.85 ± 14.02	yes	5.70 ± 0.46	55.4 ± 25.6
AP-50	0.757 ± 0.178	127.75 ± 22.88	yes	5.94 ± 0.24	69.7 ± 20.5
AP-100	0.681 ± 0.187	174.98 ± 31.03	yes	6.00 ± 0.00	76.1 ± 11.4
AP-200	0.542 ± 0.143	243.16 ± 39.01	yes	6.00 ± 0.00	75.1 ± 8.4
AP-500	0.451 ± 0.111	373.61 ± 46.36	yes	6.00 ± 0.00	72.2 ± 6.1
AP-1000	0.363 ± 0.060	544.91 ± 53.25	yes	6.00 ± 0.00	68.1 ± 3.7
AP-2000	0.259 ± 0.006	855.98 ± 3.45	yes	6.00 ± 0.00	60.1 ± 0.2
AP-5000	0.254 ± 0.003	891.66 ± 5.37	yes	6.00 ± 0.00	61.5 ± 0.2
<i>adversarial</i>	0.246 ± 0.000	969.55 ± 0.00	yes	6.00 ± 0.00	60.8 ± 0.0

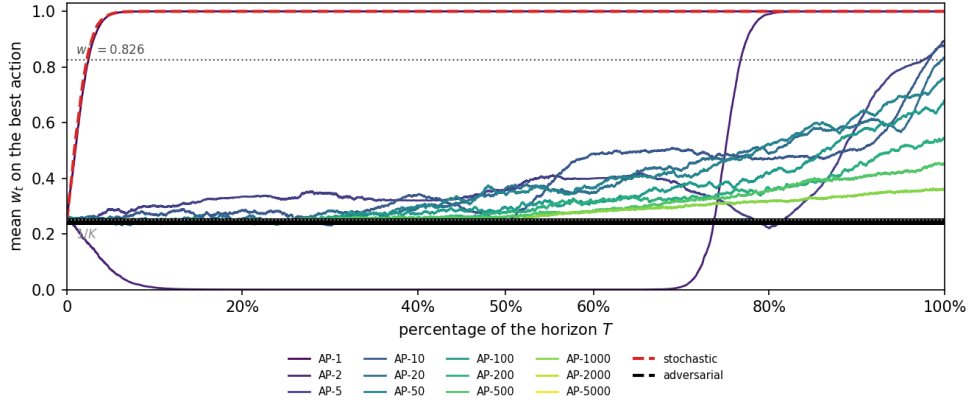


Figure 1: The weight on the overall best action, w_t under Hedge($2\eta^*$), averaged over 50 runs, against the fraction of the horizon elapsed. This is the quantity Lemma 15 uses to bound δ_t and Lemma 16 requires to converge. AP-1 lies on top of **stochastic**: both reach $w^* = 1$ inside the first 1% of the horizon, which is why they appear as a wall at the left edge. The settings with many short windows lie on top of **adversarial**, and the rest fan out between them in order of n , most of them doing their work in the last third. AP-2 is the curve that falls and then jumps back at 75%. With two windows, the hindsight-best action is the one favoured in the *second* window. For the first half of the horizon it is therefore the worse action.

Write $\rho = \frac{1}{T} \sum_t (\max_k \ell_t^k - \min_k \ell_t^k)$ for the average disagreement between the actions *within* a round, and let the *spread* be the final range $\max_k L_T^k - \min_k L_T^k$, which measures how far the totals drift apart *across* the horizon. The construction holds the first fixed and varies the second:

Table 3: Along the whole sweep the within-round disagreement ρ is constant, so it cannot be what changes. The spread falls by three orders of magnitude, and $\Delta_T(1)$ follows it. Every action carries the same number of losses in every window, so narrowing the windows tightens the constraint and nothing else moves.

	AP-1	AP-10	AP-100	AP-1000	AP-5000
within-round ρ	0.875	0.875	0.875	0.875	0.876
spread	2000	291	81	17	1
$\Delta_T(1)$	1.77	39.50	174.98	544.91	891.66
w_T^*	1.000	0.878	0.681	0.363	0.254

Every action carries exactly $\lfloor W/2 \rfloor$ losses in every window, so the totals can never drift further apart than the windows allow. Raising n shortens the windows and tightens that constraint, and does nothing else: ρ does not move, the spread collapses, and the mixability gap rises with it. This is Lemma 15 read from the unfavourable side. When the totals separate, the weight concentrates and the bound $(e-2)\eta(1-w_t^*)$ drives δ_t towards zero; when an exact count forbids them from separating, the weight stays near $1/K$ and the bound stops helping.

But is the data really harder? A natural objection is worth answering. If all four totals end up equal, surely it does not matter which action you pick, so the setting should be easy, not hard. The objection is right, and the numbers agree with it.

	R of constant $1/K$	R of AdaHedge	Hedge's $\mathbf{w}_t \cdot \boldsymbol{\ell}_t$	uniform	difference
AP-1	1500.0	3.9	0.300	0.450	-0.150
AP-10	127.7	34.9	0.464	0.473	-0.009
AP-100	43.2	76.9	0.488	0.476	+0.013
AP-1000	8.4	67.9	0.541	0.487	+0.053
AP-5000	0.8	61.5	0.589	0.500	+0.089
adversarial	0.0	60.8	0.347	0.250	+0.097

A learner that plays the uniform weight $1/K$ every round and never changes it has regret 0.8 on AP-5000 and exactly 0 on **adversarial**. These settings are not hard. They are trivial for an algorithm that does nothing.

So Δ_T does not measure how hard the data is. It measures the price of Hedge's adaptivity, which is what Definition 3 says it is, and under an exact per-window count that price becomes a loss. The counts are fixed in advance, so an action that has taken few of its ones early in a window must take more of them later. Hedge moves weight onto an action exactly when that action has been lucky, which is exactly when its luck is about to run out. The last three columns show the result: at AP-1 the weights help, paying 0.300 per round against the uniform 0.450, but from AP-100 onwards they cost between 0.013 and 0.097 per round, and over 10^4 rounds that is the whole regret.

Going down the sweep, then, the data does not become harder in any absolute sense. It becomes data on which adapting is the wrong thing to do. An independent control confirms the reading. Fix $n = 10$, so $W = 1000$, and vary only *who* loses the x_i ones, keeping the schedule and hence the total of 18900 losses identical: taking them from one fixed action for the whole run gives spread 1100, $w_T^* = 1.000$ and $\Delta_T(1) = 13.86$; taking them from a new leader each window is AP-10 itself, with spread 291 and 39.50; taking them evenly from all four gives spread 0, $w_T^* = 0.252$ and 222.52. The mixability gap moves by a factor of 16 while the amount of loss does not move at all.

Remark 23 (A flat posterior is not enough on its own). Lemma 15 is an upper bound, so a flat posterior makes it useless but does not on its own make δ_t large. Both ingredients are needed: the posterior must be spread *and* the losses must disagree within the round. The extreme case is immediate from Definition 3: if every action took loss 1 in every round, the weight would sit at $1/K$ for the whole horizon, yet the centred loss $\ell_t - \mathbf{w}_t \cdot \boldsymbol{\ell}_t$ would be identically zero and $\delta_t = 0$ exactly. The two columns of Table 3 are therefore both necessary. **adversarial** has $\rho = 1$ and reaches 969; AP-5000 has $\rho = 0.876$ and reaches 892; **stochastic** has the same $\rho = 0.875$ as the whole sweep but $w_T^* = 1$, and reaches 1.54. Neither condition alone predicts anything.

4.4 The crossover point

Section 1 claimed no single fixed rate serves both ends. With twelve settings that becomes a curve, and the curve crosses.

Table 4: Final regret of the three fixed rates, mean $\pm$ sd over 50 runs, in the order of Table 2. The aggressive rate degrades from 24.1 to 62.8 and the cautious one improves from 92.4 to 15.6.

regime	Hedge $2\eta^*$	Hedge η^*	Hedge $\eta^*/2$
<i>stochastic</i>	24.0 ± 0.3	46.8 ± 0.4	92.4 ± 0.6
AP-1	24.1 ± 0.2	46.9 ± 0.1	92.4 ± 0.1
AP-2	30.2 ± 0.5	56.4 ± 0.3	108.9 ± 0.2
AP-5	38.8 ± 5.7	60.8 ± 10.8	97.4 ± 21.6
AP-10	49.5 ± 5.1	61.2 ± 11.0	81.7 ± 23.9
<i>low-gap</i>	50.0 ± 6.0	59.6 ± 8.4	69.0 ± 18.4
AP-20	54.9 ± 5.3	60.1 ± 10.4	70.7 ± 19.5
AP-50	57.8 ± 4.9	54.7 ± 10.2	55.3 ± 16.9
AP-100	59.3 ± 5.1	49.4 ± 10.5	44.6 ± 14.6
AP-200	57.5 ± 4.7	41.7 ± 7.8	33.4 ± 9.6
AP-500	55.4 ± 4.5	36.0 ± 6.2	25.9 ± 7.0
AP-1000	51.9 ± 3.1	30.4 ± 3.7	19.6 ± 4.0
AP-2000	48.2 ± 0.2	24.5 ± 0.1	12.6 ± 0.0
AP-5000	44.9 ± 0.3	22.8 ± 0.1	11.8 ± 0.1
<i>adversarial</i>	62.8 ± 0.0	31.3 ± 0.0	15.6 ± 0.0

To find where they trade places we need a *paired* test, because both rates run on exactly the same data. Write $d_s = R_s(\eta^*/2) - R_s(2\eta^*)$ for the difference on run s . The mean of d , with its number of standard errors in brackets, is $+32.2$ (10.7) at AP-10, $+15.8$ (6.5) at AP-20, -2.5 (1.2) at AP-50 and -14.7 (9.7) at AP-100. The sign flips between AP-20 and AP-50. But AP-50 itself sits at only 1.2 standard errors, so its sign is not clear. What the data supports is a range, not a point: the difference is clearly positive at AP-20 and clearly negative at AP-100, so the crossover is somewhere between those two, with AP-50 inside the range rather than marking it. The range also contains *low-gap* ($+19.1$ at 6.2 standard errors), which is not part of the family. So the crossover is not something we designed into AP.

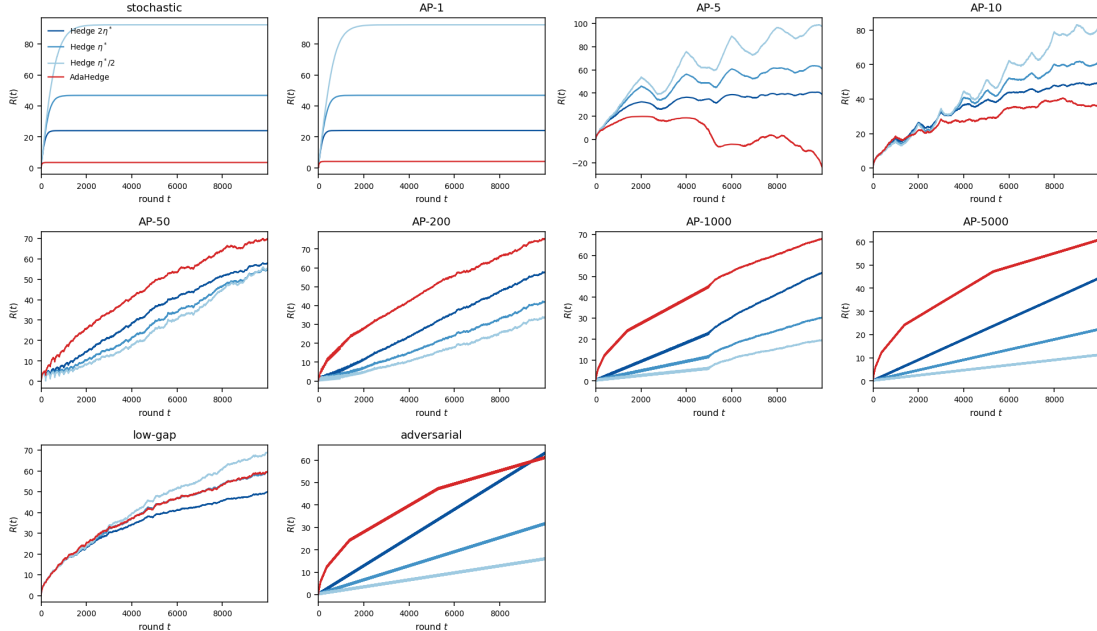


Figure 2: Regret against time, mean of 50 runs, with $\min_k L_t^k$ recalculated every round, for a selection of the settings. Across the range the light curve (Hedge $\eta^*/2$) starts worst and ends best while the dark one does the reverse. The flat AdaHedge curve in *stochastic* and AP-1 is Lemma 17 in action: one segment, η never lowered. The corners elsewhere are restarts, each re-paying $\ln K$ at a halved rate, which is the geometric cost of Lemma 12.

4.5 Are the proven bounds satisfied?

Lemma 12 and Theorem 13 were checked on every individual run, not on the averages, and both hold on all of them. The Lemma 12 bound is 3.7, 10.2, 22.2, 45.3, 90.6, 180.3 at $m = 1, \dots, 6$, and the comparison must be made run by run, because each run has its own m . Done that way: on **stochastic** the 42 runs with $m = 1$ face a bound of 3.7 and the largest regret among them is 3.52, so the worst case reaches 95% of its own bound; over all 50 runs the mean ratio $R/\text{bound}(m)$ is 0.70 and the maximum 0.95. On **adversarial**, where every run has $m = 6$ and a bound of 180.3, the ratio is 0.34.

One row needs a separate comment. AP-5 has mean regret -23.9 , which means that on most runs AdaHedge finishes *below* every fixed action. None of the bounds in Section 3 forbid this, because they are all upper bounds. But it does show the baseline failing.

Two things have to happen together for this, and $n = 5$ is where they overlap. The first is that the baseline must be weak. Write G for the amount by which an algorithm that always followed the current window leader would beat the best fixed action. The second is that AdaHedge must actually be able to follow that leader, which depends on how large the per-window gap x_i is against the within-window noise $\sqrt{W}$.

n	G	$x_n/\sqrt{W}$	weight on the current leader	runs with $R < 0$
1	0	20.0	1.000	0%
2	500	14.1	0.991	2%
5	640	8.9	0.824	48%
10	682	6.3	0.565	12%
20	692	4.5	0.420	2%

The two quantities move in opposite directions. G rises with n , because more switches make the fixed baseline weaker, and it is zero at $n = 1$ where the leader never changes. The weight on the current leader falls with n , because shorter windows carry smaller gaps against the same noise. At $n = 2$ AdaHedge follows almost perfectly but there is little to gain; at $n = 10$ there is more to gain but it holds only 0.565 of its weight on the right action. Only at $n = 5$ are both large at once. The standard deviation ± 69.4 is worth noting too: the regret is negative on 48% of runs, so this is closer to a coin flip than to a property of the setting, and which way a run falls depends on whether one action happened to lead the windows with the largest x_i .

The right baseline for data with a moving leader is the best *sequence* of actions with a limited number of changes, which is called tracking regret. AdaHedge, as analysed here, has no guarantee against it.

4.6 The constant hidden in $O(K + \log(1/\delta))$

Theorem 19 hides how big its constant really is. Evaluate (12) through Lemma 12 for **stochastic**, where $K = 4$, the gap is $0.5 - 0.3 = 0.2$ so $\alpha = 0.1$, and $\delta = 0.05$: this gives $m^* = 9$ and a regret ceiling of 1425. The measured regret is 3.5, so the guarantee is 407 times larger than reality. Worse, the worst-case bound of Theorem 13 gives 341 on the same data, so *the constant-regret theorem is the weaker of the two guarantees at this horizon*. It becomes the stronger one only once $3.46\sqrt{\frac{4}{e-1}}L_T^* \ln K$ becomes bigger than 1425, which at a loss rate of 0.3 needs $T \approx 1.75 \times 10^5$.

The cause is the dependence on α . The second term of (12) carries $\alpha^{-2} \ln(1/\alpha^2)$, so m^* grows like $\log_\phi(\alpha^{-2})$ and the bound, being geometric in m^* , grows like α^{-2} . For **low-gap**, where $\alpha = 0.005$, this gives $m^* = 18$ and a ceiling of 7.3×10^5 , which is useless because $R(T) \leq T = 10^4$ holds anyway. The theorem says nothing about that setting at this horizon, and the measured 59.7 must have another explanation: the signal $0.01 \cdot T = 100$ and the noise $\sqrt{T} = 100$ are the same size, so the best action is almost impossible to identify, even in theory. That is a limit of statistics, not of the algorithm.

4.7 Where the improved upper bound does not help

Lemma 15 beats Lemma 8 only when $(e - 2)\eta(1 - w_t^*) < \eta/8$, that is when

$$1 - w_t^* < \frac{1}{8(e - 2)} \approx 0.174, \quad \text{i.e.} \quad w_t^* > 0.826.$$

Below that point the improved bound is the worse one, so the easy-data analysis does nothing until the weight has *already* concentrated. It explains why concentration keeps going once it starts, but not how it starts.

The sweep crosses this line, which is part of why it was built. AP-1 up to AP-20 sit above it, from 1.000 to 0.836; AP-50 at 0.757 is the first below, and the rest run down to 0.254. In eight of the twelve settings Lemma 15 is strictly worse than the basic Lemma 8, so the easy-data argument does not apply at all, even though the data plainly has a favoured action in every window. The threshold falls at almost exactly the same place as the crossover of Section 4.4, between AP-20 and AP-50; nothing in the analysis predicts that, and with two events and twelve settings it may well be a coincidence.

4.8 A structural limit shown by the proof

Lemma 8 has a consequence the paper does not draw. Because $\delta_t \leq \eta/8$, using up the budget takes at least

$$T_{\min}(\eta) = \frac{b(\eta)}{\eta/8} = \frac{8 \ln K}{\eta^2} + \frac{8 \ln K}{(e-1)\eta} \quad (14)$$

rounds whatever the data is. With $K = 4$ and $\eta_i = \phi^{1-i}$ the first five segments need

$$T_{\min} = 17.5, 57.3, 203.3, 761.4, 2942.4,$$

so the earliest possible restart rounds are

$$17.5, 74.8, 278.1, 1039.5, 3981.9.$$

This sequence is fixed by ϕ alone. The doubling trick makes AdaHedge quadratically slower to restart as η drops, so by design it cannot react to anything that happens late in the run.

The sweep measures how close the data can push it to that floor. The mean restart rounds are

$$63, 218, 701, 2755 \quad \text{at } n = 10, \quad 25, 104, 376, 1391, 5304 \quad \text{at } n = 5000.$$

The second set is within a factor of 1.43 of the minimum at every segment. So the hardest setting this family can build is one where AdaHedge restarts as early as the mixability gap allows, and no earlier.

Note — *Reading this as a limit, not a failure.* AdaHedge is not built to spot changing patterns, and it does not claim to be. What (14) shows is that the doubling trick makes it quadratically slower to restart as η falls, and by Note 4.2 it cannot restart more than six times at $T = 10^4$ whatever the data does. Both facts come from the choice of ϕ , not from the budget rule. The successor paper [3] removes them by making the monitored quantity part of the definition, $\eta_t = \ln K / \Delta_{t-1}$, so that η falls smoothly instead of in steps and the weights are never reset. The limitation shown here is exactly the one that change is designed to fix.

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